D. Color Cooccurrence Experiment

Here we describe the details of how we created the four color representations visualized in Figure 8, from left to right.

Perceptual representation from CIELAB color space We embed pixels taken from the CIFAR-10 image dataset (Krizhevsky et al., 2009; Torralba et al., 2008) based on the CIELAB color space, which is designed as a *perceptually uniform* space that changes numerical values correspond to similar perceived changes in color.

Three representations from cooccurrence in VISION and LANGUAGE For these three representations, we first obtain a dissimilarity matrix over colors (in different ways detailed below), then use multidimensional scaling (Shepard, 1980) to find a 3-dimensional embedding in which Euclidean distance between the embeddings for A and B, z_A and z_B, best matches this dissimilarity matrix. We use 1,000 fits and take the best match. Afterward, we visually align it with the CIELAB space by finding the best rotation, translation, scaling, and flipping, by running the Kabsch-Umeyama algorithm (Kabsch, 1976; 1978; Umeyama, 1991) twice, once on $\mathbf{z}$ and once on $-\mathbf{z}$, to account for flipping. The dissimilarity matrix we used in each case is described as following:

- **VISION: Pixel cooccurrence.** We collect color cooccurrence statistics from the CIFAR-10 dataset, and estimate a joint distribution $p(A, B)$ over 300,000 randomly sampled pixel colors A and B that occur within a radius of at most 4 pixels of one another. Colors are quantized on a grid in RGB space and represented as discrete variables, and $p(A, B)$ is modeled as a table of normalized counts, from which we compute the empirical pointwise mutual information matrix $K_{\mathsf{PMI}}(A, B)$. Quantization ensures that there is no bias from how color distances are represented in RGB space. Dissimilarity matrix is defined as $-K_{\mathsf{PMI}}(A, B) + c$, where $c = \max_{A,B} K_{\mathsf{PMI}}(A, B)$ is an offset to ensure non-negativity (similar to the constant in Section 4.2 and Proposition F.1 that ensures neural networks can express K_{PMI}).

- **LANGUAGE.** We used an approach similar to Abdou et al. (2021).
  - We take 20 pairs of (color, word) appeared in the dataset collected by Lindsey & Brown (2014), where 51 participants were asked to free name each of the 330 colors from the Munsell Color Chart. We filtered words that appeared less than 100 times, and computed each word's associate color by taking the centroid in CIELAB space. Our filtering process followed Abdou et al. (2021) exactly, but resulted in 20 colors, a slightly different set than the 18 colors they claimed.
  - For each of the 20 color words `<col>`, we construct three sentences:

    ```
    The color <col>.
    This color is <col>.
    The color of this thing is <col>.
    ```

    and obtain the average sentence embedding from the language encoder, as the embedding for `<col>` (details below). We find this approach more effective than Abdou et al. (2021), which uses object names that potentially have color biases, even though the objects may appear in multiple colors.
  - Unlike Abdou et al. (2021), we did not perform linear regression from language embedding to CIELAB space, which distorts distances and easily overfits with only 20 samples. Instead, we used multidimensional scaling to best preserve distances, as described above.
  - **Masked language contrastive learning (SimCSE) embedding:** We used sentence embedding from the unsupervised SimCSE RoBERTa-L (Gao et al., 2021) to encode the above sentences into 1024-dimensional embeddings, and used the pairwise Euclidean distances among `<col>` embeddings as the dissimilarity matrix.
  - **Masked language predictive learning (RoBERTa) embedding:** We concatenated hidden states of the last four layers of RoBERTa-L (Liu et al., 2019), following (Devlin et al., 2018). We averaged across token dimensions, and obtained a 4096-dimensional embedding for each of the above sentences, and used the pairwise Euclidean distances among `<col>` embeddings as the dissimilarity matrix.

E. Caption Density Experiments

We use LLaMA3-8B-Instruct (Meta, 2024) to generate summary captions at various densities for images in the Densely Captioned Images dataset (Urbanek et al., 2023) from the train split. Following Urbanek et al. (2023), we prompt the language model with the following instructions to generate captions at differing granularity:

```
system: You are given a full-text description of an image. You should summarize it into about
<num_words> words, being sure to include as much salient visual information as possible given the
<num_words> word constraint, especially information from the start of the original description. The
new description should apply for the original image. Respond with only the summary, in one line.

user: <original_caption>
```

We measure the alignment with this generated caption to test our hypothesis that denser captations would result in higher alignment scores. In Figure 9, we find that the alignment score also improves as caption length increases.

F. Analysis of Contrastive Learners

F.1. Contrastive objectives learn pointwise mutual information

There are two widely used forms of contrastive objectives. We now discuss each form in detail and show how they both are minimized by the pointwise mutual information (PMI) as stated in Equation (5). To simplify notation, we consider learning the bivariate model $g(x_a, x_b) \in \mathbb{R}$. In Section 4, such g is optimized within the family of $\{g = \langle f_X, f_X \rangle \colon f_X \in \mathcal{F}_X\}$.

Recall that our positive pairs are sampled from $(x, x_+) \sim P_{\mathsf{coor}}$, and that the negative pairs are sampled independently from its marginals which we denote as $(x, x_-) \overset{\text{i.i.d.}}{\sim} P$ where $P(x) = \sum_{x_+} P_{\mathsf{coor}}(x, x_+)$.

1. **The binary NCE loss (Gutmann & Hyvärinen, 2010)** is defined with a certain prior over sampling positive vs. negative pairs. Let p_{pos} be the probability of sampling a positive pair. Then the loss is given by

$$\mathcal{L}_{\mathsf{binary\text{-}NCE}}(g) \triangleq p_{\mathsf{pos}} \cdot \mathbb{E}_{(x,x_+)\sim P_{\mathsf{coor}}} \left[-\log \sigma(g(x, x_+))\right] + (1 - p_{\mathsf{pos}}) \cdot \mathbb{E}_{(x,x_-)\overset{\text{i.i.d.}}{\sim} P} \left[-\log \sigma(-g(x, x_-))\right]. \tag{20}$$

The Bayes optimal solution is given by

$$\begin{aligned} g(x_a, x_b) &= \log \frac{P(\mathsf{pos} \mid x_a, x_b)}{1 - P(\mathsf{pos} \mid x_a, x_b)} && (21)\\ &= \log \frac{P(\mathsf{pos}, x_a, x_b)}{P(\mathsf{neg}, x_a, x_b)} && (22)\\ &= \log \frac{p_{\mathsf{pos}} \cdot P_{\mathsf{coor}}(x_a, x_b)}{(1 - p_{\mathsf{pos}}) P(x_a) P(x_b)} && (23)\\ &= \log \frac{P_{\mathsf{coor}}(x_a, x_b)}{P(x_a) P(x_b)} + \log \frac{p_{\mathsf{pos}}}{1 - p_{\mathsf{pos}}} && (24)\\ &= K_{\mathsf{PMI}}(x_a, x_b) + c_X. && (25) \end{aligned}$$

2. **The InfoNCE loss (Oord et al., 2018)** is defined with randomly sampling one positive pair along with K negative ones. With some hyperparameter $\tau > 0$, the loss is given by

$$\mathcal{L}_{\mathsf{InfoNCE}}(g) \triangleq \mathbb{E}_{\substack{(x,x_+)\sim P_{\mathsf{coor}} \\ (x_-^{(1)}, x_-^{(2)}, \ldots, x_-^{(K)}) \overset{\text{i.i.d.}}{\sim} P}} \left[-\log \frac{e^{g(x,x_+)/\tau}}{e^{g(x,x_+)/\tau} + \sum_{i=1}^{K} e^{g(x,x_-^{(i)})/\tau}} \right]. \tag{26}$$

The Bayes optimal solution is given by

$$\begin{aligned} \frac{e^{g(x,x_+)/\tau}}{e^{g(x,x_+)/\tau} + \sum_{i=1}^{K} e^{g(x,x_-^{(i)})/\tau}} &= \frac{P_{\mathsf{coor}}(x_+ \mid x) \prod_j P(x_-^{(j)})}{P_{\mathsf{coor}}(x_+ \mid x) \prod_j P(x_-^{(j)}) + \sum_i P_{\mathsf{coor}}(x_-^{(i)} \mid x) P(x_+) \prod_{j \neq i} P(x_-^{(j)})} && (27)\\ &= \frac{P_{\mathsf{coor}}(x_+ \mid x)/P(x_+)}{P_{\mathsf{coor}}(x_+ \mid x)/P(x_+) + \sum_i P_{\mathsf{coor}}(x_-^{(i)} \mid x)/P(x_-^{(i)})}. && (28) \end{aligned}$$

For $\tau = 1$, this optima corresponds to g choices where

$$g(x_a, x_b) = \log \frac{P_{\mathsf{coor}}(x_b \mid x_a)}{P(x_b)} + c_X(x_a) \tag{29}$$

$$= K_{\mathsf{PMI}}(x_a, x_b) + c_X(x_a). \tag{30}$$

For the general $\tau \neq 1$ case, we have g (and corresponding f_X) recovers K_{PMI} up to an offset and a scale. Our main argument in Section 4 that f_X recovers K_{PMI} still holds.

F.2. Contrastive learners can represent K_{PMI} exactly under smoothness conditions

We want to express $K_{\mathsf{PMI}} + C$ using some representation function $f_X \colon \mathcal{X} \to \mathbb{R}^n$ so that

$$\langle f_X(x_a), f_X(x_b) \rangle = K_{\mathsf{PMI}}(x_a, x_b) + C, \qquad \text{for some } C. \tag{31}$$

For such an f_X to exist, an equivalent criterion is that $K_{\mathsf{PMI}} + C$ is positive semi-definite (PSD), as can be seen from eigendecomposition.

Proposition F.1. *Suppose that the off-diagonal elements of K_{PMI} are bounded within $[\log \rho_{\mathsf{min}}, \log \rho_{\mathsf{min}} + \delta] \in (-\infty, 0]$. We have $K_{\mathsf{PMI}} + C$ is positive semi-definite (PSD) for some C if the joint distribution is sufficiently smooth:*

$$\frac{P_{\mathsf{coor}}(z_i \mid z_i)}{P_{\mathsf{coor}}(z_i)} \geq e^{N\delta} \rho_{\mathsf{min}}, \qquad \forall i. \tag{32}$$

Proof. Note that $K_{\mathsf{PMI}} + C$ still only has non-positive off-diagonal elements if

$$-C \geq \log \rho_{\mathsf{min}} + \delta. \tag{33}$$

For such C, it is diagonally dominant (and thus PSD) if,

$$\forall i, \qquad K_{\mathsf{PMI}}(z_i, z_i) + C \geq \sum_{j \neq i} |K_{\mathsf{PMI}}(z_i, z_j) + C| = -(N-1)C - \sum_{j \neq i} K_{\mathsf{PMI}}(z_i, z_j), \tag{34}$$

or equivalently,

$$\forall i, \qquad NC + \sum_j K_{\mathsf{PMI}}(z_i, z_j) \geq 0. \tag{35}$$

The following choice of C readily satisfies the above Equation (35):

$$C \triangleq -\min_i \frac{1}{N} \sum_j K_{\mathsf{PMI}}(z_i, z_j). \tag{36}$$

Therefore, it remains to show that Equation (33) is true. Note that

$$-C \triangleq \min_i \frac{1}{N} \sum_j K_{\mathsf{PMI}}(z_i, z_j) \geq \frac{N-1}{N} \log \rho_{\mathsf{min}} + \frac{1}{N} (\min_i K_{\mathsf{PMI}}(z_i, z_i)). \tag{37}$$

Therefore, it suffices to have

$$\log \rho_{\mathsf{min}} + \delta \leq \frac{N-1}{N} \log \rho_{\mathsf{min}} + \frac{1}{N} (\min_i K_{\mathsf{PMI}}(z_i, z_i)). \tag{38}$$

Rearranging terms gives the desired condition

$$\frac{P_{\mathsf{coor}}(z_i \mid z_i)}{P_{\mathsf{coor}}(z_i)} \geq e^{N\delta} \rho_{\mathsf{min}}, \qquad \forall i. \tag{39}$$

□

Remark F.2. Proposition F.1 is one example that a sufficiently smooth world or a sufficiently high sampling rate allows the PMI kernel K_{PMI} to be *exactly* represented as inner products of a learned feature space (up to a scale). The condition here can be satisfied, for example, if the off-diagonal terms decay linearly with respect to N and stay sufficiently close to each other. While the condition is somewhat strict, it captures the essence that smoothness and continuity allow easier learning. Nonetheless, we note that exact representation is not necessary for convergence, and thus this requirement can likely be relaxed. Please see Section 6 for discussions on practical settings.